REPUBLIC OF KENYA

Kenya Institute of Curriculum Development — Grade 10

MATHEMATICS

LEARNER'S PAMPHLET

A Quick-Reference Guide to Every Topic in Grade 10 Mathematics

🎓 **WHO IS THIS FOR?**

This pamphlet is for YOU — a Grade 10 learner in Kenya's Senior School. It breaks down every topic in the Mathematics curriculum into simple explanations, key formulas, worked examples, and real Kenyan-life connections, so you can revise with confidence at school or at home.

Strands covered: Numbers and Algebra • Measurements and Geometry • Statistics and Probability

Table of Contents

How to Use This Pamphlet

Each topic in this pamphlet follows the same easy structure:

Key Ideas: The core facts and rules you must know.

📐 **Formula Box:** The exact formulas to memorise and apply.

Worked Example: A step-by-step solved problem showing the method.

🌍 **Why It Matters in Kenya:** A real-life connection to help the idea stick.

✍️ **Try It Yourself:** A quick practice question — attempt it before checking with your teacher.

STRAND 1: NUMBERS AND ALGEBRA

1.1 Real Numbers *(8 lessons)*

Every number you use — from counting cows to reading your phone's data balance — belongs to a family of numbers. This topic helps you sort numbers into groups and find their reciprocals (their 'flip' value).

Key Ideas

- Whole numbers can be odd, even, prime, or composite.
- Real numbers split into rational numbers (can be written as a fraction, e.g. 1/2, 0.75, 4) and irrational numbers (cannot be written as an exact fraction, e.g. $\sqrt{2}$, π).
- The reciprocal of a number is 1 divided by that number. The reciprocal of a/b is b/a.
- Reciprocals can be found by division, using mathematical tables, or a calculator.

📐 **FORMULA BOX**

Reciprocal of a number n = 1 ÷ n

Reciprocal of a fraction a/b = b/a

Worked Example

Find the reciprocal of 0.8.

0.8 = 4/5, so its reciprocal = 5/4 = 1.25.

🌍 **WHY IT MATTERS IN KENYA**

When farmers share a bag of fertiliser equally, or a matatu sacco splits daily fare collections among 8 vehicles, reciprocals help calculate each share quickly and accurately.

✍ **TRY IT YOURSELF**

Classify 17 and 24 as prime/composite, then find the reciprocal of 5/8.

1.2 Indices and Logarithms *(8 lessons)*

Indices (powers) let us write repeated multiplication in short form, while logarithms let us 'undo' powers. Together they make big or tiny numbers easy to work with.

Key Ideas

- Index form: a^n means a multiplied by itself n times.
- Laws of indices: $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$.
- A logarithm to base 10 answers the question: '10 raised to what power gives this number?'
- $\log_{10}(a \times b) = \log_{10}a + \log_{10}b$, and $\log_{10}(a \div b) = \log_{10}a - \log_{10}b$.

📐 **FORMULA BOX**

If $10^x = N$, then $\log_{10}N = x$

$\log(ab) = \log a + \log b$ | $\log(a/b) = \log a - \log b$ | $\log(a^n) = n \log a$

Worked Example

Simplify $2^3 \times 2^4 \div 2^2$.

$= 2^{3+4-2} = 2^5 = 32$.

🌍 **WHY IT MATTERS IN KENYA**

Logarithms are used to read the Richter scale for earthquakes, calculate compound interest on a SACCO loan, and measure sound levels (decibels) at a stadium.

✍ **TRY IT YOURSELF**

Simplify $3^5 \times 3^2 \div 3^4$, then find $\log_{10}100$.

1.3 Quadratic Expressions and Equations 1 *(14 lessons)*

A quadratic expression has a squared term (like x^2). Learning to form, factorise, and solve these equations helps you model curved or 'turning-point' situations in real life.

Key Ideas

- A quadratic expression has the form $ax^2 + bx + c$.
- Common identities: $(a+b)^2 = a^2 + 2ab + b^2$; $(a-b)^2 = a^2 - 2ab + b^2$; $(a+b)(a-b) = a^2 - b^2$.
- Factorising means writing a quadratic as a product of two brackets.
- To solve by factorisation: set the equation to zero, factorise, then let each bracket equal zero.

📐 **FORMULA BOX**

$ax^2 + bx + c = 0 \rightarrow$ factorise to $(x - p)(x - q) = 0 \rightarrow x = p$ or $x = q$

Worked Example

Solve $x^2 - 5x + 6 = 0$.

Factorise: $(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$.

🌍 **WHY IT MATTERS IN KENYA**

A farmer with a rectangular shamba wants to know what length and width give a required area — this is solved using a quadratic equation. Quadratics also model the path of a ball kicked during a football match.

✍ **TRY IT YOURSELF**

Factorise $x^2 + 7x + 12$, then solve $x^2 - x - 6 = 0$.

STRAND 2: MEASUREMENTS AND GEOMETRY

2.1 Similarity and Enlargement *(12 lessons)*

Similar shapes have the same form but different sizes. Enlargement is the transformation that produces similar shapes using a centre point and a scale factor.

Key Ideas

- Linear Scale Factor (L.S.F) = length of image ÷ corresponding length of object.
- Area Scale Factor (A.S.F) = $(L.S.F)^2$.
- Volume Scale Factor (V.S.F) = $(L.S.F)^3$.
- A negative scale factor produces an image on the opposite side of the centre of enlargement.

📐 **FORMULA BOX**

A.S.F = $(L.S.F)^2$

V.S.F = $(L.S.F)^3$

Worked Example

A model car has L.S.F = 1/20 of the real car. If the real car's bonnet has an area of 2 m^2, find the model's bonnet area.

A.S.F = $(1/20)^2$ = 1/400. Model area = 2 ÷ 400 = 0.005 m^2.

🌍 **WHY IT MATTERS IN KENYA**

Architects use enlargement to make scaled building plans; a map of Kenya is a huge enlargement (reduction) of the actual country; passport photo studios enlarge or reduce printed photos.

✍ **TRY IT YOURSELF**

Two similar water tanks have a linear scale factor of 3. If the smaller tank holds 50 litres, find the capacity of the larger tank.

2.2 Reflection and Congruence *(12 lessons)*

A reflection creates a mirror image of a shape across a line. Congruent shapes are identical in size and shape — reflection is one way of producing them.

Key Ideas

- A line of symmetry divides a shape into two mirror-image halves.
- In a reflection, the object and image are equidistant from the mirror line, and the line joining a point to its image is perpendicular to the mirror line.
- Congruence tests for triangles: SSS, SAS, AAS/ASA, and RHS.

📐 **FORMULA BOX**

Object point and image point are equidistant from the mirror line, and the mirror line is the perpendicular bisector of the line joining them.

Worked Example

Reflect point A(2, 3) in the line $y = x$.

Reflecting in $y = x$ swaps the coordinates: image A' = (3, 2).

🌍 **WHY IT MATTERS IN KENYA**

The Kenyan flag design, kiondo weaving patterns, and building facades often use reflective symmetry. Side mirrors on matatus and boda bodas use the principles of reflection.

✍ **TRY IT YOURSELF**

Reflect the point B(−1, 4) in the x-axis, then state which congruence test applies when two triangles share three equal sides.

2.3 Rotation *(10 lessons)*

Rotation turns a shape about a fixed point (the centre of rotation) through a given angle, without changing its size.

Key Ideas

- A rotation is described fully by its centre and angle (with direction — clockwise or anticlockwise).
- Order of rotational symmetry = number of times a shape fits onto itself in one full turn (360°).
- Rotation produces congruent (direct congruent) images.

📐 **FORMULA BOX**

Order of rotational symmetry = 360° ÷ smallest angle that maps the shape onto itself

Worked Example

A regular hexagon is rotated about its centre. Find its order of rotational symmetry.

$360° \div 60° = 6$. The hexagon has rotational symmetry of order 6.

🌍 **WHY IT MATTERS IN KENYA**

Bicycle and matatu wheels, a spinning ceiling fan, and traditional Kenyan basket (kiondo) patterns are everyday examples of rotation and rotational symmetry.

✍ **TRY IT YOURSELF**

State the order of rotational symmetry of a square, then describe how you would rotate a point (2, 0) by 90° anticlockwise about the origin.

2.4 Trigonometry 1 *(15 lessons)*

Trigonometry studies the relationship between the angles and sides of right-angled triangles — a powerful tool for measuring things you cannot reach directly, like the height of a tree.

Key Ideas

- SOH-CAH-TOA: $\sin\theta$ = opposite/hypotenuse, $\cos\theta$ = adjacent/hypotenuse, $\tan\theta$ = opposite/adjacent.
- $\sin\theta = \cos(90° - \theta)$ — sines and cosines of complementary angles are equal.
- Special angles 30°, 45°, 60°, 90° have exact trigonometric ratios derived from an equilateral and an isosceles right-angled triangle.
- Angle of elevation is measured upward from the horizontal; angle of depression is measured downward.

📐 **FORMULA BOX**

$\sin\theta$ = Opp/Hyp

$\cos\theta$ = Adj/Hyp

$\tan\theta$ = Opp/Adj = $\sin\theta / \cos\theta$

Worked Example

A learner stands 20 m from a flagpole and measures the angle of elevation to the top as 30°. Find the height of the flagpole.

$\tan 30° = \text{height}/20 \rightarrow \text{height} = 20 \times \tan 30° = 20 \times 0.577 = 11.5$ m.

🌍 **WHY IT MATTERS IN KENYA**

Surveyors use trigonometry to measure the height of buildings and hills (like the slopes of Mt. Kenya); pilots use it for navigation; it's also used to work out the correct pitch of an iron-sheet roof.

✍ **TRY IT YOURSELF**

A ladder leans against a wall, making a 60° angle with the ground. If the ladder is 4 m long, find how high up the wall it reaches.

2.5 Area of Polygons *(14 lessons)*

Beyond the basic base × height formula, there are smarter ways to find the area of triangles and many-sided shapes — useful whenever land or materials need to be measured.

Key Ideas

- When two sides and the included angle of a triangle are known, use the sine formula for area.
- Heron's formula finds a triangle's area using only its three side lengths.
- Regular polygons (pentagon, hexagon, heptagon, octagon...) can be split into equal triangles from the centre to find total area.

📐 **FORMULA BOX**

Area = $\frac{1}{2}ab\sin C$

Heron's formula: Area = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = (a+b+c)/2$

Worked Example

A triangular plot has sides 7 m and 9 m with an included angle of 40°. Find its area.

Area = $\frac{1}{2} \times 7 \times 9 \times \sin 40° = 31.5 \times 0.643 = 20.25$ m² (2 d.p.).

> 🌍 **WHY IT MATTERS IN KENYA**
>
> Surveyors and farmers use these formulas to calculate the area of irregular shambas for buying, selling, or leasing land, and for estimating quantities of seed or fertiliser needed.

> ✍️ **TRY IT YOURSELF**
>
> A triangle has sides 5 cm, 6 cm and 7 cm. Use Heron's formula to find its area.

2.6 Area of a Part of a Circle *(12 lessons)*

Circles rarely appear whole in real problems — often we need just a slice (sector), a ring (annulus), or a curved strip (segment) of a circle.

Key Ideas

- An annulus is the region between two concentric circles (a ring shape).
- A sector is a 'pizza-slice' region bounded by two radii and an arc.
- A segment is the region between a chord and an arc.
- The area of an annular sector combines the sector and annulus ideas.

> 📐 **FORMULA BOX**
>
> Area of sector = $(\theta/360°) \times \pi r^2$
>
> Area of annulus = $\pi(R^2 - r^2)$
>
> Area of segment = Area of sector − Area of triangle

Worked Example

Find the area of a sector of radius 14 cm and angle 90°. (Use $\pi = 22/7$)

Area = $(90/360) \times 22/7 \times 14^2 = 0.25 \times 22/7 \times 196 = 154$ cm².

> 🌍 **WHY IT MATTERS IN KENYA**
>
> A car's windscreen wiper sweeps out a sector; a dartboard is made of sectors and annuli; a chapati or pizza slice models a sector; irrigation sprinklers water a sector-shaped area of a farm.

> ✍️ **TRY IT YOURSELF**
>
> A circular garden of radius 10 m has a path of width 2 m running around its outer edge. Find the area of the path (the annulus).

2.7 Surface Area and Volume of Solids *(15 lessons)*

This topic shows you how to 'wrap' (surface area) and 'fill' (volume) 3D objects — from a simple box to a water tank shaped like a cylinder or a sphere.

Key Ideas

- Surface area is the total area of all outer faces of a solid; volume is the amount of space it occupies.
- Composite solids are made of two or more basic solids joined together — find each part's area/volume separately, then combine.
- The volume of a cone is exactly one-third the volume of a cylinder with the same base and height.

> 📐 **FORMULA BOX**
>
> Cylinder: Volume = $\pi r^2 h$, Curved S.A. = $2\pi rh$
>
> Cone: Volume = $(1/3)\pi r^2 h$, Curved S.A. = πrl
>
> Sphere: Volume = $(4/3)\pi r^3$, S.A. = $4\pi r^2$

Worked Example

A cylindrical water tank has radius 1.4 m and height 2 m. Find its volume. (Use $\pi = 22/7$)

Volume = $22/7 \times 1.4^2 \times 2 = 22/7 \times 1.96 \times 2 = 12.32$ m³.

> 🌍 **WHY IT MATTERS IN KENYA**
>
> Water tanks (cylinders), grain silos, footballs (spheres), and packaging boxes (cuboids/composite solids) are everyday objects whose material cost and capacity depend on these formulas.

> ✍️ **TRY IT YOURSELF**
>
> Find the volume of a cone with base radius 3 cm and height 7 cm. (Use $\pi = 22/7$)

2.8 Vectors I *(15 lessons)*

A vector describes a quantity with both size (magnitude) and direction, unlike a scalar which has size only. Vectors let us describe movement and forces precisely.

Key Ideas

- A vector can be written as a column vector (x, y) or using notation like AB or a bold letter.
- Equivalent vectors have the same magnitude and direction (they can start at different points).
- Vectors add 'tip-to-tail'; multiplying a vector by a scalar changes its size (and direction if the scalar is negative).
- Magnitude of vector (x, y) is found using Pythagoras' theorem: $|v| = \sqrt{x^2 + y^2}$.

> 📐 **FORMULA BOX**
>
> Vector addition: $(a,b) + (c,d) = (a+c, b+d)$
>
> Magnitude: $|v| = \sqrt{x^2 + y^2}$
>
> Midpoint of vector AB = $((x_1+x_2)/2, (y_1+y_2)/2)$

Worked Example

Find the magnitude of vector v = (3, 4).

$|v| = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

🌍 **WHY IT MATTERS IN KENYA**

Pilots and ship captains use vectors to combine speed and direction (e.g. accounting for wind or ocean currents); a boda boda rider's displacement from home to town is a vector.

✍️ **TRY IT YOURSELF**

Given a = (2, 5) and b = (−1, 3), find a + b and the magnitude of a.

2.9 Linear Motion *(15 lessons)*

This topic connects mathematics to everyday movement — how far, how fast, and how quickly speed changes — using graphs to picture a journey.

Key Ideas

- Distance is the total path covered; displacement is the straight-line distance from start to finish, in a specific direction.
- Speed = distance/time; velocity = displacement/time (speed in a stated direction).
- Acceleration is the rate of change of velocity.
- On a displacement-time graph, the gradient gives velocity. On a velocity-time graph, the gradient gives acceleration and the area under the graph gives distance.

📐 **FORMULA BOX**

Speed = Distance ÷ Time

Acceleration = (Final velocity − Initial velocity) ÷ Time

Relative speed (same direction) = difference of speeds; (opposite directions) = sum of speeds

Worked Example

A matatu travels 90 km in 1.5 hours. Find its average speed.

Speed = 90 ÷ 1.5 = 60 km/h.

🌍 **WHY IT MATTERS IN KENYA**

Understanding speed and relative speed is vital for road safety when overtaking, and Kenya's world-class athletes and coaches use velocity-time graphs to analyse race performance and improve training.

✍️ **TRY IT YOURSELF**

Two cars are 300 m apart, moving towards each other at 20 m/s and 25 m/s. Find how long it takes for them to meet.

STRAND 3: STATISTICS AND PROBABILITY

3.1 Statistics I *(16 lessons)*

Statistics is about collecting, organising, and making sense of data so that we can make good decisions — an essential skill for business, government, and everyday life.

Key Ideas

- Data can be grouped into classes and displayed in a frequency distribution table.
- Mean is the average; median is the middle value; mode is the most frequent value.
- Histograms and frequency polygons are visual tools to represent grouped data.
- Reading and interpreting a graph correctly is as important as drawing it.

📐 **FORMULA BOX**

Mean = $\Sigma(fx) \div \Sigma f$ (for grouped data, x = midpoint of each class)

Median = value at position $(n+1)/2$ in ordered data (ungrouped)

Worked Example

Find the mean of the data set: 4, 6, 6, 8, 10, 12.

Sum = 46, number of values = 6. Mean = 46 ÷ 6 = 7.67 (2 d.p.).

🌍 **WHY IT MATTERS IN KENYA**

The Kenya National Bureau of Statistics uses these exact tools to analyse census data, rainfall patterns, and market prices, helping the government plan for schools, hospitals, and food security.

✍ **TRY IT YOURSELF**

The marks of 5 students in a CAT are: 45, 60, 55, 60, 70. Find the mean, mode, and median.

3.2 Probability 1 *(12 lessons)*

Probability measures how likely an event is to happen, ranging from impossible (0) to certain (1). It helps us make informed decisions when outcomes are uncertain.

Key Ideas

- Experimental probability = number of times an event occurs ÷ total number of trials.
- Probability of any event lies between 0 and 1 inclusive.
- Mutually exclusive events cannot happen at the same time; independent events do not affect each other's outcome.
- Addition law is used for 'or' situations; multiplication law is used for 'and' situations. Tree diagrams organise combined outcomes clearly.

📐 **FORMULA BOX**

P(event) = Number of favourable outcomes ÷ Total possible outcomes

Mutually exclusive: P(A or B) = P(A) + P(B)
Independent events: P(A and B) = P(A) × P(B)

Worked Example

A die is tossed once. Find the probability of getting a number greater than 4.

Favourable outcomes: {5, 6} → 2 outcomes. P = 2/6 = 1/3.

🌍 **WHY IT MATTERS IN KENYA**

Meteorologists use probability to forecast rainfall (important for farmers planning planting season); insurance and betting companies use it to set fair prices; it also helps in games like bao and ludo.

✍ **TRY IT YOURSELF**

A bag contains 4 red and 6 blue balls. A ball is picked at random. Find the probability that it is red, then find the probability of picking two red balls in a row (without replacement).

A Word of Encouragement

Mathematics is a skill built through daily practice, not talent alone. Every topic in this pamphlet connects to real situations you already see around you — in the market, on the road, on the farm, and even on the football pitch. Revisit the 'Try It Yourself' questions regularly, work with classmates, ask your teacher when stuck, and keep a positive mindset. You have what it takes to master Grade 10 Mathematics!

💡 **REMEMBER**

Practice a little every day rather than a lot once in a while.

Show your working — it helps you (and your teacher) find and fix mistakes.

Use digital tools like calculators and educational apps, but understand the method first.